

Silicon-based phase shifters for high figure of merit in optical modulation

Kensuke Ogawa^{*a}, Kazuhiro Goi^a, Norihiro Ishikura^a, Hiroki Ishihara^a, Shinichi Sakamoto^a, Tsung-Yang Liow^b, Xiaoguang Tu^b, Guo-Qiang Lo^b, Dim-Lee Kwong^b, Soon Thor Lim^c, Min Jie Sun^d, and Ching Eng Png^c

^aAdvanced Technology Laboratory, Fujikura Ltd., 1440 Mutsuzaki, Sakura, Chiba, 285-8550, Japan;

^bInstitute of Microelectronics, 11 Science Park Road, Singapore Science Park II, 117685, Singapore;

^cInstitute of High-Performance Computing, 1 Fusionopolis Way, #16-16 Connexis, 138632, Singapore; ^dOptic2Connect Pte. Ltd., 71 Ayer Rajah Crescent, #03-17, 139951, Singapore

ABSTRACT

This paper focuses on latest progress in experimental and theoretical studies on silicon-based carrier-depletion PN-junction phase shifters in terms of high modulation efficiency for energy-efficient photonic networks of high-transmission capacity. Modulation efficiency of rib-waveguide phase shifters having various PN-junction configuration are characterized with respect to DC figure of merit defined for phase shifters using carrier-plasma dispersion as the physical principle of refractive-index modulation. In addition, RF drive voltage required for 10-Gb/s on-off keying is characterized for rib-waveguide phase shifters including lateral and vertical PN-junction configurations.

Keywords: optical modulator, silicon photonics, photonic integrated circuit, figure of merit, PN junction

1. INTRODUCTION

Silicon Mach-Zehnder modulators (MZMs) are suitable for high-contrast optical modulation in broad spectral ranges with symbol rates as high as 50 Gbaud.¹ Silicon MZMs attract extensive attention for applications to low-cost small-footprint optical transceivers and active optical cables in energy-efficient optical networks in telecom, datacom and on-chip optical-interconnect applications.²⁻¹¹ In telecom applications, in particular, high-speed optical modulation beyond 100 Gb/s/λ for optical-fiber transmission in advanced modulation formats such as dual-polarization quadrature phase-shift keying (DP-QPSK) has been achieved by using lateral PN-junction carrier-depletion silicon MZMs, which were monolithically integrated with optical circuits such as waveguides for polarization-division multiplexing (PDM) on a silicon-on-insulator (SOI) wafer in small chip footprints.²⁻⁵ Optical-fiber transmission up to 1000 km in 128-Gb/s DP-QPSK format has been demonstrated by using the monolithic silicon PDM IQ MZM copackaged with electrical modulator drivers in a ceramic-based metal package in a small form factor of 15x35 mm² in footprint and 4.5 mm in profile.^{4,12} Peak-to-peak RF voltage (RF V_{pp}) of ~6.5 V or higher in 50-ohm impedance matching produces 180-degree phase shift in the silicon MZM, and further reduction in the drive voltage is required for the deployment of the compact silicon modulator in energy-efficient optical transport networks.

This paper focuses on latest progress in experimental and theoretical studies on silicon rib-waveguide carrier-depletion phase shifters having various PN-junction configurations in the light of high modulation efficiency for energy-efficient photonic networks of high-transmission capacity. The rib-waveguide phase shifters are suitable for monolithic integration with optical circuits such as PDM waveguides under common design rule in CMOS-friendly fabrication processes. Modulation efficiency of rib-waveguide phase shifters are characterized with respect to DC figure of merit (FOM), which is defined for phase shifter operating on the basis of carrier-plasma dispersion in refractive-index modulation. In theoretical analysis, DC FOM is obtained from effective refractive-index change under DC reverse bias voltages using numerical model of two-dimensional carrier distribution in PN-junction phase shifters. In addition, RF V_{pp} required for 10-Gb/s on-off keying (OOK) with extinction ratio higher than 10 dB is presented and characterized for phase shifters including one with vertical PN-junction configuration.

*kensuke.ogawa@jp.fujikura.com

2. CARRIER-DEPLETION RIB-WAVEGUIDE PHASE SHIFTER

Silicon MZM having a lateral PN-junction rib-waveguide phase shifters is illustrated in Fig. 1. The Si MZM consists of single MZ interferometer (MZI) and RF coplanar waveguide (CPW) on the MZI. Lateral PN junction was formed in Si rib-waveguide phase shifters with length $\ell = 3$ or 4 mm in the both arms of the MZI. The rib waveguide was designed with the dimensions of 500-nm rib width, 220-nm rib height and 95-nm side slab height for optical modulation in single-mode transverse-electric polarization. Vertical boundary of PN junction is positioned in the middle of the rib waveguide. The dimensions were determined to achieve high modulation speed and low optical loss.^{13,14} Higher slab provides lower series resistance and faster phase modulation, whereas optical loss is higher due to weaker mode confinement in the center rib area with the slab height higher than 95 nm. The Si MZMs were fabricated on 8-inch SOI wafers of 220-nm SOI thickness and 2- μm buried-oxide (BOX) thickness using 130-nm node CMOS processes. The BOX layer serves as the bottom clad. DC reverse-bias voltage is applied to PN-junction in each phase shifter for high-speed refractive-index modulation in carrier-depletion mode.

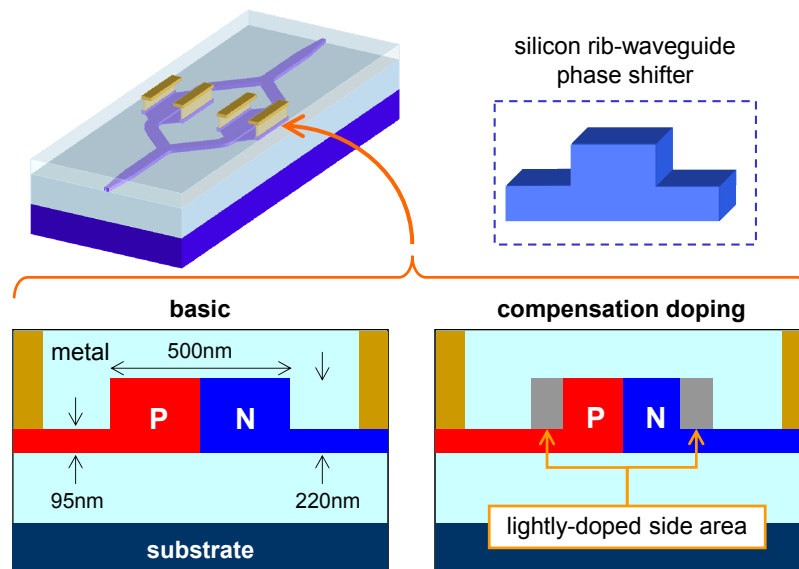


Figure 1. Silicon MZM with rib-waveguide phase shifters of basic and compensation-doping lateral PN-junction profiles.

Capacitive coupling of reverse-biased PN junction was included in RF simulation of CPW to obtain a signal electrode of 10- μm width and gaps of 6.4- μm width between signal and ground electrodes for 50- Ω impedance matching. The RF electrodes were made of Al metal deposited on 2- μm thick top clad in 2- μm thickness. The signal electrode was connected to P-doped slab of the rib waveguide, and one of the ground electrodes to N-doped slab of the rib waveguide through Al vertical interconnect access. The CPW was designed using short and straight signal electrodes without bending. The signal electrodes were deposited only on top of the phase shifters and connected with contact pads disposed on the both sides. The signal and ground electrodes were wire bonded to external electrodes on a package.¹²

Two profiles of carrier-depletion Si MZMs having lateral PN-junction in rib-waveguide phase shifter were prepared as shown in Fig. 1 for DC FOM analysis. Rib-waveguide phase shifter in one of the profiles has basic PN-junction profile, in which left half of the silicon core in the rib waveguide was uniformly doped with P-type dopants (boron) and right half of the core was doped uniformly with N-type dopants. Depletion region is formed along the vertical boundary of the PN-junction at the middle of the rib core. Lightly doped side areas were formed in the other profile of compensation-doping PN-junction. The side area of P-type region of the left half of the core was counter doped with N-type dopants, and the other side area of the N-type region was counter doped with P-type dopants to reduce carrier-induced optical loss.¹⁵ The compensation doping allow reduction of optical loss in PN-junction phase shifter without causing increase in π -shift voltage (V_π), thereby enabling enhancement of modulation efficiency.

3. FIGURE OF MERIT ANALYSIS

3.1 DC optical characteristics

Experimental and theoretical characteristics of DC FOM were derived from DC reverse-bias voltage dependences of phase shift and carrier-induced optical loss. Experimental DC reverse-bias voltage dependences of phase shift and carrier-induced optical loss were obtained by optical transmission measurements for the Si MZMs with swept DC reverse-bias voltage.⁴ For theoretical characteristics, two-dimensional device simulation based on a finite-element solver for charge continuity and carrier transport equation combined with Poisson equation was performed to obtain numerical carrier distribution profiles in the lateral PN-junction phase shifters under DC reverse-bias voltages.¹⁶ Electron and hole concentrations in N- and P-type areas are $2 \times 10^{17} \text{ cm}^{-3}$ and $5 \times 10^{17} \text{ cm}^{-3}$, respectively. Carrier distribution profiles were converted to refractive-index profiles in respective PN-junction phase shifters using the empirical formula on the free-carrier plasma dispersion.¹⁷ Phase shift and carrier-induced optical loss were derived after light-propagation simulation in the PN-junction phase shifters using the simulated refractive-index profile by beam propagation method.¹⁸ Measured and phase shift and carrier-induced optical loss for the basic PN-junction profile are plotted in Fig. 2 and coincide well with the simulated characteristics. Carrier-induced loss in Fig. 2 provides phase shifter optical loss, which includes not only optical loss due to free-carrier absorption but also passive optical loss of undoped rib waveguide $\sim 0.07 \text{ dB/mm}$. Measured and simulated phase shift and carrier-induced optical loss were obtained also for a rib-waveguide phase shifter with compensation-doping profile.¹⁵

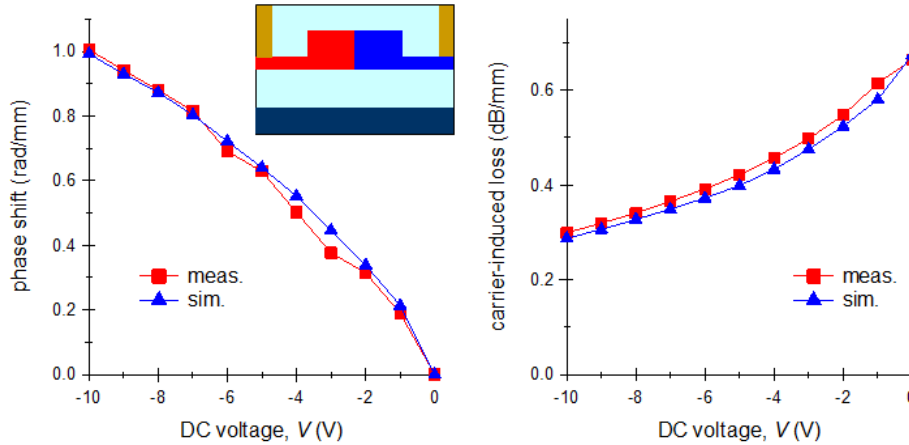


Figure 2. Measured and simulated phase shift and carrier-induced optical loss with DC reverse-bias voltages for phase shifter with basic PN-junction profile.

3.2 DC FOM of phase shifters with basic and compensation-doping PN junction

DC FOM in this study represents π phase shift against phase shifter optical loss (α_{PS}) and π -shift voltage (V_{π}) with the following equation:

$$\text{FOM} = \frac{\pi}{\alpha_{PS} V_{\pi}} \quad (1)$$

DC FOM of the phase shifters with basic and compensation-doping PN-junction profiles plotted versus Dc reverse-bias voltage in Fig. 3 were obtained from the measured phase shift and carrier-induced optical loss as presented in Fig. 2. Higher DC FOM was obtained at DC reverse-bias voltages below 5 V for the compensation-doping PN junction than for the basic PN-junction since residual free-carrier absorption is suppressed in the lightly doped side areas as described in the previous section.¹⁵ DC FOM for the compensation-doping PN-junction profile is slightly lower, however, at DC reverse-bias voltages above 5 V. This is because the depletion region extends and overlaps with the light doped side areas at stronger DC reverse-bias voltage, namely more negative DC voltage, and refractive index change is weaker. Therefore, higher modulation efficiency with the compensation-doping PN-junction profile is obtained under DC reverse-bias voltage $> -5 \text{ V}$

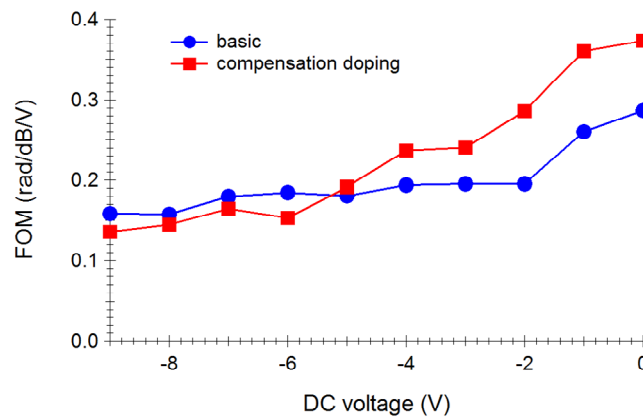


Figure 3. DC FOM vs DC reverse-bias voltage for rib-waveguide phase shifters with basic and compensation-doping PN-junction profiles.

3.3 Further enhancement with serpentine PN-junction configuration

Further enhancement of DC FOM is expected with serpentine PN-junction configuration presented in Fig. 4. In this configuration, PN-junction is folded and phase-shifter length is effectively extended along the light propagation direction. The configuration is similar to interleaved or interdigitated PN-junction phase shifters in terms of folded PN junction layout.¹⁹⁻²⁴ The serpentine PN-junction presented here is, however, more suitable to higher modulation efficiency than with the conventional interleaved PN junction because lightly doped side area has been formed in N-doped area of the rib core.

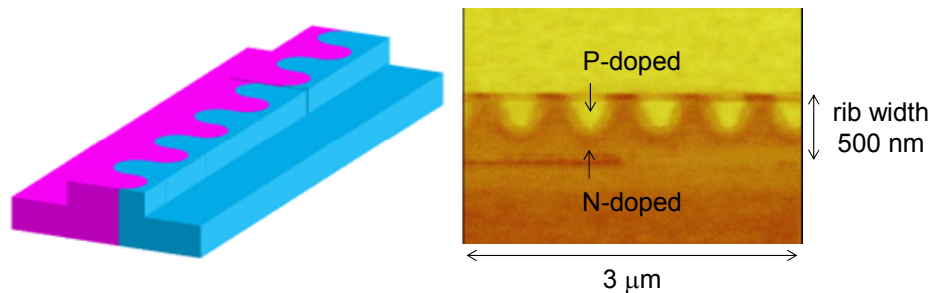


Figure 4. Illustrated rib-waveguide phase shifter with serpentine PN-junction configuration (left) and top-view scanning capacitance microscopy image of P and N dopant potential distribution in rib-waveguide phase shifter .

Formation of serpentine PN-junction shape was confirmed experimentally as microscopy images obtained by means of scanning capacitance microscopy (SCM). The imaging method was developed for potential mapping of submicron-scale junction transistors on the basis of atomic force microscopy.^{25,26} Top-view SCM image of dopant potential distribution is presented in Fig. 4. Serpentine PN junction is clearly observed in the image, in which dopant potential distribution is revealed as color contrast.

DC FOM of the rib-waveguide phase shifter with the serpentine PN-junction configuration was obtained in simulation with $\ell = 3$ mm as plotted in Fig. 5. Concentrations of P- and N-type dopants were changed between $1 \times 10^{17} \text{ cm}^{-3}$ and $1 \times 10^{18} \text{ cm}^{-3}$. Significant enhancement of DC FOM is predicted in the numerical analysis: twice as large as that for the basic PN-junction configuration with linear junction layout along the propagation direction at a DC reverse-bias voltage of -3 V if dopant concentration is optimized. Simulated DC V_{π} is 3.6 V or lower, which leads to $V_{\pi}\ell_{\pi}$ of 1.08 V·cm or lower.

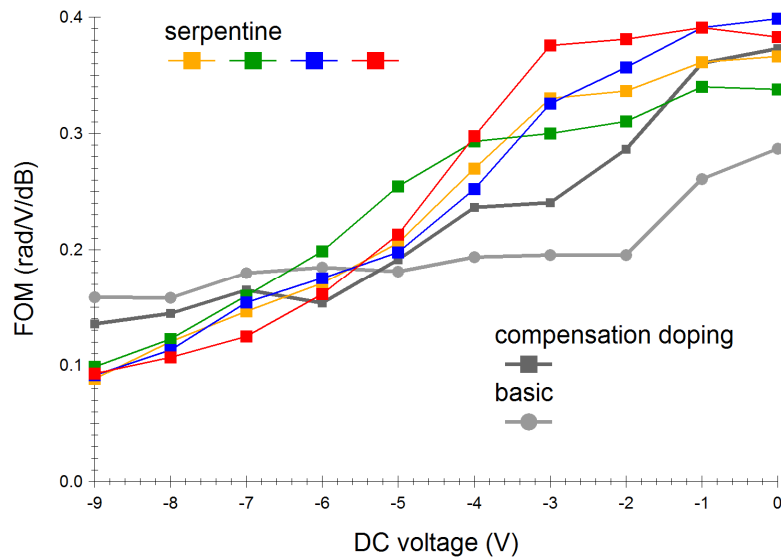


Figure 5. Simulated DC FOM with serpentine PN-junction configuration.

4. HIGH-SPEED MODULATION CHARACTERISTICS

High-speed modulation performances of the Si MZMs having 4-mm rib-wave guide phase shifters in the lateral PN-junction configurations were characterized in 10-Gb/s OOK eye-diagram measurements. For basic PN-junction configuration with linear junction shape along the propagation direction, RF V_{pp} was about 8 V with extinction ratio higher than 13 dB as shown in Fig. 6. RF V_{pp} was reduced to ~6.5 V with the phase shifter in the serpentine PN-junction configuration. Modulation voltage reduction has been achieved with design modification using the serpentine configuration.

Further remarkable reduction in RF V_{pp} , as low as 3.3 V_{pp} with 11-dB extinction ratio, has been confirmed for Si MZM having 3-mm vertical PN-junction phase shifters. Profile of the vertical PN junction is illustrated in Fig. 6.²⁷ DC FOM is derived as ~0.25 with 2.7-V DC V_{π} . The DC FOM in the vertical PN junction is lower than the DC FOM simulated at a DC reverse-bias voltage of -3 V in the serpentine configuration. The vertical PN junction configuration has the advantage of reduction in modulation voltage.

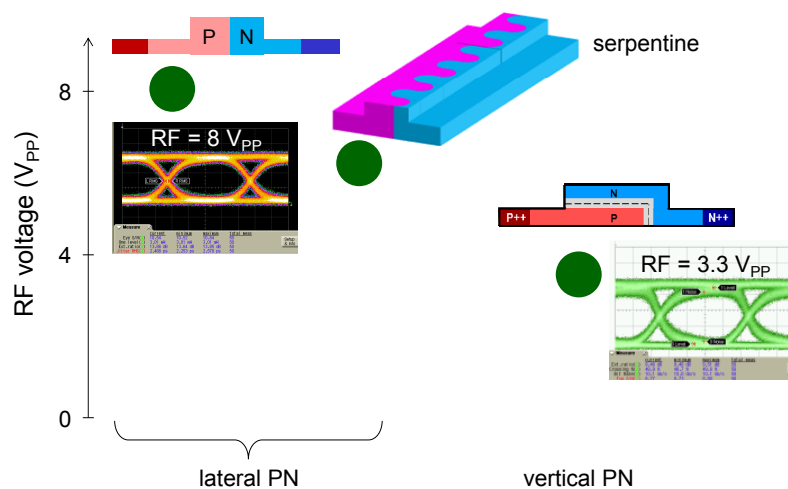


Figure 6. Layout of fiber link and measurement setup for transmission measurement, Si PDM IQ modulator assembled in a package, measured constellation diagrams in two orthogonal linear polarization states.

5. CONCLUSION

Experimental and theoretical studies on silicon rib-waveguide carrier-depletion phase shifters having various PN-junction configurations have been reviewed in the light of high modulation efficiency for energy-efficient photonic networks of high-transmission capacity. Modulation efficiency of the rib-waveguide phase shifters are characterized with respect to DC FOM, which represents π phase shift against phase shifter optical loss and π -shift voltage. Compensation-doping PN-junction profile, in which lightly doped side areas in rib-waveguide core allow optical loss reduction due to free-carrier absorption, thereby higher DC FOM. Further enhancement of DC FOM is predicted with the serpentine PN-junction layout. High-speed performances of Si MZMs in the lateral and vertical PN-junction configurations were characterized in 10-Gb/s OOK eye-diagram measurements. RF V_{PP} as low as 3.3 V_{PP} was achieved with 11-dB extinction ratio for the vertical PN-junction phase shifter.

REFERENCES

- [1] Reed, G. T., Thomson, D. J., Gardes, F. Y., Hu Y., Fedeli, J.-M. and Mashanovich, G. Z., "High-speed carrier-depletion silicon Mach-Zehnder optical modulators with lateral PN junctions," *Front. Phys.* 2, 77 (2014).
- [2] Dong, P., Xie, C., Chen, L., Buhl, L. L., and Chen, Y.-K., "112-Gb/s monolithic PDM-QPSK modulator in silicon," *Opt. Express* 20(26), B624-B629 (2012).
- [3] Doerr, C., Chen, L., Vermeulen, D., Nielsen, T., Azemati, S., Stulz, S., McBrien, G., Xu, X.-M., Mikkelsen, B., Givchchi, M., Rasmussen, C., and Park, S.-Y., "Single-chip silicon photonics 100-Gb/s coherent transceiver," *Th5C.1*.
- [4] Ogawa, K., Ishihara, H., Goi, K., Mashiko, Y., Lim, S. T., Sun, M. J., Seah, S., Png, C. E., Liow, T.-Y., Tu, X., Lo, G.-Q., and Kwong, D.-L., "Fundamental characteristics and high-speed applications of carrier-depletion silicon Mach-Zehnder modulators," *IEICE Electronics Express* 11(24), 20142010 (2014).
- [5] Ogawa, K., Goi, K., Oka, A., Mashiko, Y., Liow, T.-Y., Tu, X., Lo, G.-Q., Kwong, D.-L., Lim, S. T., Sun, M. J., and Png, C. E., "Design and characterization of high-speed monolithic silicon modulators for digital coherent communication," *Proc. SPIE* 9367, 93670C-1-93670C-8 (2015).
- [6] De Dobbelaere, P., Abdalla, S., Gloeckner, S., Mack, M., Masini, G., Mekis, A., Pinguet, T., Sahni, S., Guckenberger, D., Harrison, M., and Adithyaram Narasimha A., "Si photonics based high-speed optical transceivers," 38th European Conference and Exhibition on Optical Communication (ECOC, 2012) We.1.E.5.
- [7] Webster, M., Appel, C., Gothoskar, P., Sunder, S., Dama, B., and Shastri, K., "Silicon photonic modulator based on a MOS-capacitor and a CMOS driver," *IEEE Compound Semiconductor Integrated Circuit Symposium (CSICs)*, 2014) E.3.
- [8] Kai, Y., Nishihara, M., Tanaka, T., Okabe, R., Takahara, T., Rasmussen, J. C., Ishihara, H., Goi, K., and Ogawa, K., "130-Gbps DMT transmission using silicon Mach-Zehnder modulator with chirp control at 1.55- μ m," *Optical Fiber Communication Conference and Exposition (OFC)*, 2015) Th.4A.1.
- [9] Gill, D. G., Proesel, J. E., Xiong, C., Orcutt, J. S., Rosenberg, J. C., Khater, M. H., Barwicz, T., Assefa, S., Shank, S. M., Reinholm, C., Ellis-Monaghan, J., Kiewra, E., Kamlapurkar, S., Breslin C. M., Green, W. M. J., Haensch, W., and Vlasov, Y. A., "Demonstration of a high extinction ratio monolithic CMOS integrated nanophotonic transmitter and 16 Gb/s optical link," *IEEE J. Sel. Top. Quantum Electron.* 21(4), 3400311 (2015).
- [10] Baba, T., Akiyama, S., Imai, M., and Usuki, T., "25-Gb/s broadband silicon modulator with 0.31-V $\cdot$ cm V_{π} L based on forward-biased PIN diodes embedded with passive equalizer," *Opt. Express* 23(26), 32950-32960 (2015).
- [11] Sun, C., Wade, M. T., Lee, Y., Orcutt, J. S., Alloatti, L., Georgas, M. S., Waterman, A. S., Shainline, J. M., Avizienis, R. R., Lin, S., Moss, B. R., Kumar, R., Pavanello, F., Atabaki, A. H., Cook, H. M., Ou, A. J., Leu, J. C., Chen, Y.-H., Asanović, K., Ram, R. J., Popović, M. A., and Stojanović, V. M., "Single-chip microprocessor that communicates directly using light," *Nature* 528, 534–538 (2015).

- [12] Goi, K., Oda, K., Mashiko, Y., Ogawa, K., Liow, T.-Y., Tu, X., Lo, G.-Q., and D.-L. Kwong “128Gb/s DP-QPSK silicon modulator module integrated with driver amplifiers,” 40th European Conference on Optical Communications (ECOC, 2014), P.2.20.
- [13] Ogawa, K., Goi, K., Tan, Y. T., Liow, T.-Y., Tu, X., Fang, Q., Lo, G.-Q. and Kwong D.-L., “Silicon Mach-Zehnder modulator of extinction ratio beyond 10 dB at 10.0-12.5 Gbps,” *Opt. Express* 19(26), B26-B31 (2011).
- [14] Ogawa, K., Goi, K., Kusaka, Terada, Y., Ogawa, K., Liow, T.-Y., Tu, X., Lo, G.-Q., Kwong, D.-L., Dixit, V., Lim, S. T. and Png, C. E., “Low-loss high-speed silicon Mach-Zehnder modulator for optical-fiber telecommunications,” *Proc. SPIE* 8829, 86290U-1-86290U-8 (2013).
- [15] Tu, X., Liow, T.-Y., Song, J., Yu, M. B. and G.-Q. Lo, “Fabrication of low loss and high speed silicon optical modulator using doping compensation method,” *Opt. Exp.* 19(19), 18029-18035 (2011).
- [16] Png, C. E., Dixit, V., Lim S. T. and Li, E.-P., “Accurate high-speed eye diagram simulation of silicon-based modulators,” *Proc. SPIE* 8629 86290S-1-86290S-5 (2013).
- [17] Nedeljkovic, M., Soref, R., and Mashanovich, G. Z., “Free-carrier electrorefraction and electroabsorption modulation predictions for silicon over the 1-14- μ m infrared wavelength range,” *IEEE Photon. J.* 3(6) 1171-1180 (2011).
- [18] Png, C. E., Chan, S. P., Lim, S. T. and Reed G. T., “Optical phase modulators for MHz and GHz modulation in silicon-on-insulator (SOI),” *J. Lightwave Technol.* 22(6) 1573-1582 (2004).
- [19] Li, Z.-Y., Xu, D.-X., McKinnon, W. R., Janz, S., Schmid, J. H., Cheben, P., and Yu, J.-Z., “Silicon waveguide modulator based on carrier depletion in periodically interleaved PN junctions,” *Opt. Express* 17(18), 15947-15958 (2009).
- [20] Ziebell, M., Marris-Morini, D., Rasigade, G., Crozat, P., Fédéli, J.-M., Grosse, P., Cassan, E., and Vivien, L., “Ten Gbit/s ring resonator silicon modulator based on interdigitated PN junctions,” *Opt. Express* 19(15), 14690-14695 (2011).
- [21] Xiao, X., Xu, H., Li, X., Hu, Y., Xiong, K., Li, Z., Chu, T., Yu, Y., and Yu, J., “25 Gbit/s silicon microring modulator based on misalignment-tolerant interleaved PN junctions,” *Opt. Express* 20(3), 2507-2515 (2012).
- [22] Yu, H., Pantouvaki, M., Van Campenhout, J., Korn, D., Komorowska, K., Dumon, P., Li, Y., Verheyen, P., Absil, P., Alloatti, L., Hillerkuss, D., Leuthold, J., Baets, R., and Bogaerts W., “Performance tradeoff between lateral and interdigitated doping patterns for high speed carrier-depletion based silicon modulators,” *Opt. Express* 20(12), 12926-12938 (2012).
- [23] Rosenberg, J. C., Green, W. M. J., S. Assefa, S., Gill, D. M., T. Barwicz, T., M. Yang, M., Shank, S. M., and Vlasov, Y. A., “A 25 Gbps silicon microring modulator based on an interleaved junction,” *Opt. Express* 20(24), 26411-26423 (2012).
- [24] Goykhman, I., Desiatov, B., Ben-Ezra, S., Shappir, J., and Levy, U., “Optimization of efficiency-loss figure of merit in carrier-depletion silicon Mach-Zehnder optical modulator,” *Opt. Express* 21(17), 19518-19529 (2013).
- [25] Matey, J. R., “Scanning capacitance microscopy,” *Proc. SPIE* 0897, 110-117 (1988).
- [26] Martin, Y., Abraham, D. W., and Wickramasinghe, H. K., “High-resolution capacitance measurement and potentiometry by force microscopy,” *Appl. Phys. Lett.* 52(13), 1103-1105 (1988).
- [27] Reed, G. T., Mashanovich, G., Gardes, F. Y., Nedeljkovic, M., Hu, Y., Thomson D. J., Li, K., Wilson P. R., Chen, S.-W., and Hsu, S. S., “Recent breakthroughs in carrier depletion based silicon optical modulators,” *Nanophotonics* 3(4-5), 229-245 (2014).